

Targeted LTBI screening in the APY Lands

Updated planning report for South Australia Public Health

Research question

Does targeted screening in the APY Lands improve efficiency without compromising coverage?

This report updates the earlier planning document to reflect the current v9 analytical build: exact-age population generation, an age-based LTBI risk model, a default IGRA + 3HP pathway, do-nothing and attributable-risk add-ons, user-configurable inputs, and the backend/helper layers now being used to prepare a future browser-facing interface. That interface is under development and is not part of this report.

Prepared for

South Australia Public Health

Important use note

- This report is designed to support service planning and prioritisation, not to make decisions about any individual person.
- The model should inform testing and outreach effort. It should not be used to refuse screening or treatment.

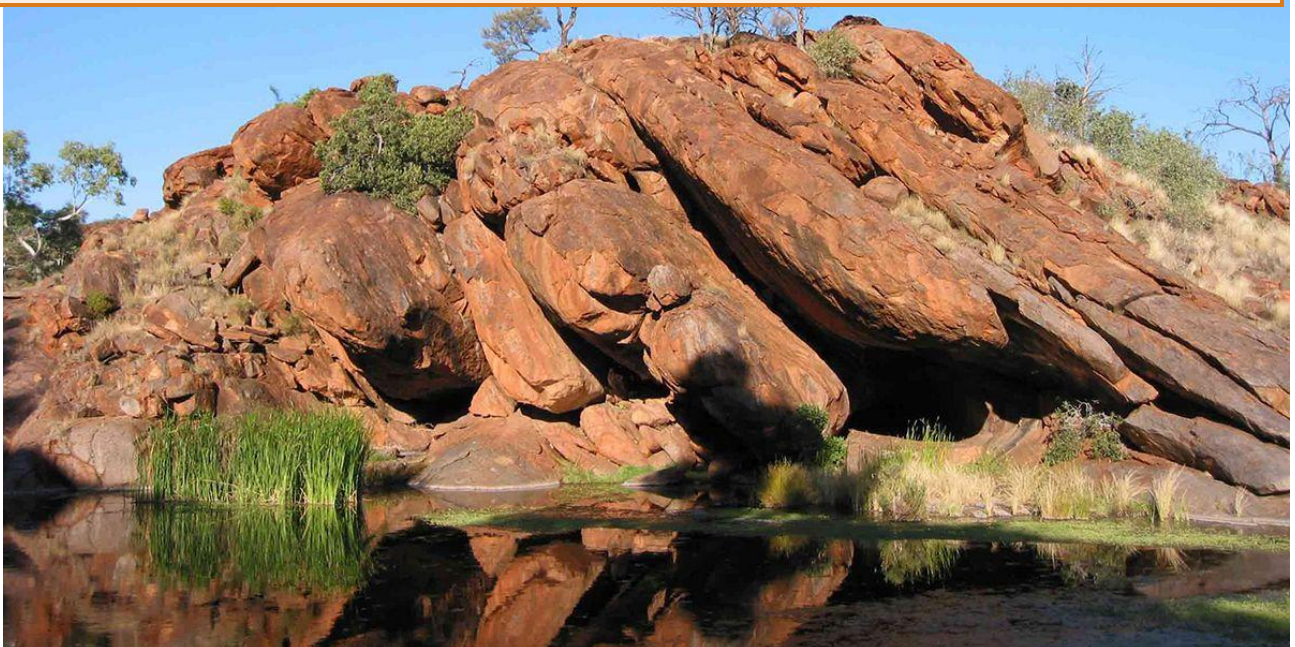


Image courtesy of [National Indigenous Times](#)

Executive summary

This report updates the earlier APY planning report to reflect the current analytical build of the MATLAB model. The current build now supports an exact-age population structure, a cumulative age-based LTBI risk model, a default IGRA + 3HP pathway, a do-nothing comparator, progression-only attributable risk outputs, a user-facing input script for local scenario testing, and backend/controller layers that are now being used to prepare a future browser-facing interface. That interface is not yet ready for deployment and is therefore not described as a deliverable in this report.

Across the illustrative IGRA + 3HP figures included here, targeted testing remains more efficient than random testing when treatment capacity is limited. The efficiency gain is largest at lower treated proportions and narrows as the treated proportion rises.

The cure-targeted strategy is more efficient than random testing for protecting or curing LTBI when only a small share of the cohort can be treated. The prevention-targeted strategy shows an even larger early advantage for preventing future active TB.

These findings support a targeted strategy for testing outreach and early treatment offer, provided it is used as a planning tool rather than a gatekeeping tool. No person should be refused screening or treatment based on this work alone.

Recommended policy position

- A targeted strategy should be pursued, weighing the efficiency gains against the convenience and operational simplicity of targeting, while ensuring that no one is refused screening or treatment on the basis of this work.

How this report should be used

- The core question is whether targeted screening improves efficiency without undermining community-wide coverage.
- The model is most useful when staff time, travel, follow-up capacity, or treatment support capacity are limited and choices must be made about where to start.
- The model is not a rule for excluding individuals. It is a testing and prioritisation tool for planning outreach, invitation, and early rollout.
- The current report uses the default IGRA + 3HP APY scenario as an illustrative example. The current model build allows users to replace many of these defaults with bespoke local inputs.

Research question and policy use

Primary question: does targeted screening in the APY Lands improve efficiency without compromising coverage?

In this report, 'efficiency' means achieving more benefit for the same programme effort - for example, curing or protecting more LTBI, preventing more future active TB, or reducing the number needed to screen or treat.

The practical use case is testing. If the programme cannot immediately offer the same intensity of outreach, follow-up, and treatment support to everyone, who should be approached first?

Important guardrail

- The model supports testing and prioritisation, not exclusion.
- Anyone who presents, is clinically indicated, or requests screening should remain able to access care.
- The advantage of targeting should be weighed against the operational convenience of simpler universal workflows.

The current analytical build is still a planning model rather than an operational protocol. It is intended to answer comparative questions such as 'what do we gain if we target early rounds of outreach?' rather than to set a rigid rule about who can or cannot be tested.

Because the model can now accept bespoke inputs, the same workflow can be reused later for other South Australian settings or for updated APY parameter sets. A browser-facing interface is now being developed on top of the same backend, but this report should still be read as the current planning and evidence document rather than as an app manual.

Methods

A closed cohort of 1,500 people is simulated and screened over the next 2 years, then followed for 20 years.

Each simulated person is assigned an exact age from the APY age distribution supplied in default_age_distribution.csv. Other risk factors remain linked to the existing three broad age groups unless the user supplies different prevalence inputs.

LTBI risk is generated using a complementary log-log form, $p(\text{LTBI}) = 1 - \exp(-\exp(\eta))$, so age behaves as a cumulative exposure process rather than a simple yes/no age-group multiplier.

How the simulation works

Step	What happens
1. Generate the cohort	Sample 1,500 people, assign exact ages from the APY age distribution, and assign risk factors using local prevalence inputs.
2. Estimate LTBI risk	Use age as a cumulative exposure process and apply the retained infection predictors for marijuana use, close contact, and renal disease.
3. Estimate untreated progression	Apply literature-based progression / reactivation risk multipliers and calibrate the active TB hazard to the observed prevalence.
4. Apply screening	Spread screening across the first 2 years, then apply the chosen test and the chosen targeting rule.
5. Apply preventive treatment	Move test-positive, non-active participants through treatment start, completion, ADR stop, and partial-treatment logic.
6. Summarise outcomes	Count false positives, treatment courses, infections protected or cured in the model, active TB cases prevented, and efficiency metrics.

Interpretation note

- The model is a person-level prevention model, not a full transmission model. Prevented TB cases in the current figures are direct person-level cases prevented, not downstream transmission effects.
- Results are strongest for comparing strategies against each other, rather than forecasting exact future case numbers.

Parameters and sources

The current report uses the default APY scenario built into the current model. These settings are illustrative and can now be replaced through the user-facing runner without changing the core model code.

Table 1. Current pathway and cascade assumptions used in the illustrative report

Parameter	Base-case value	Source / note
Illustrative cohort size per run	1,500 people	Model setting for each Monte Carlo replicate
Screening window	2 years	Current APY planning assumption
Follow-up horizon	20 years	Current APY planning assumption
Default test	IGRA	Stakeholder-selected default in v9
Default preventive regimen	3HP (12 doses over 3 months)	Stakeholder-selected default in v9
Treatment start probability	0.85	Current analytical default; user-editable
3HP completion probability	0.80	Current analytical default; user-editable
3HP treatment-limiting ADR stop	0.05	Current analytical default; user-editable
3HP full-course efficacy	0.85	Current analytical default; user-editable
Partial-course rule for 3HP	Conservative threshold at about 11/12 doses	Current analytical default; user-editable

Source note: pathway and cascade values are the current analytical defaults in the v9 build and are user-editable.

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Table 2. Calibration targets and key epidemiological inputs

Parameter	Base-case value	Source / note
LTBI prevalence target	7.53%	Barry et al. APY screening study
Active TB prevalence target	1.30%	Barry et al. APY screening study
Crude age contrast for LTBI	OR 7.54 for ≥ 25 vs < 25	Barry et al. APY screening study
Exact-age distribution	ABS-based APY age profile with smoothed column	default_age_distribution.csv supplied for this project
LTBI predictors retained in infection model	Marijuana use 3.65; close contact 2.53; renal disease 2.19	Adjusted infection ORs from Barry et al.
Progression / reactivation multipliers	Contact 5; marijuana 3; renal 3.6; diabetes 3; smoking 2; chronic lung disease 3; alcohol/drugs 3	Current literature-review / planning inputs from default_data.csv
Other risk-factor prevalences	Three broad age groups for most factors; overall values for sex and BCG	Current planning inputs from default_data.csv; user-editable in v9

Source note: Barry et al. provides the observed LTBI prevalence, active TB prevalence, and infection predictors; current planning inputs remain editable in v9.

What is new in the current build

The current analytical build is more flexible than the earlier report version. It keeps the main planning logic intact but adds several features that make the model more realistic, more useful for future scenario testing, and more ready for a future browser-facing implementation.

- Exact-age population generation from APY age-distribution data.
- Age-based cumulative LTBI risk using the complementary log-log form.
- User-facing runner (`run_tb_screening_user_options_v9.m`) with editable defaults for LTBI prevalence, age distribution, risk-factor prevalence, test choice, treatment choice, and cascade parameters.
- Do-nothing summary that reports natural-history LTBI and active TB burden against the current strategy.
- Progression-only attributable risk outputs for close contact, marijuana use, renal disease, diabetes, smoking, chronic lung disease, and harmful alcohol use.
- v9-native add-on scripts for targeting gradients, optima, profile tables, IGRA charts, do-nothing summaries, and attributable-risk charts.
- Backend, validation, results-bundle, and controller helpers are now in place to support a future browser-facing interface, although that interface is not yet part of the current deliverable.
-

Why this matters

- Exact age improves the population structure without forcing all risk factors into single-year age curves.
- The user-input runner makes the model safer to hand over because key assumptions can be changed in one place.
- Do-nothing and attributable-risk outputs make it easier to explain baseline burden and the importance of progression risk factors.

Results: NNS to cure or protect one infection

The clearest way to see how targeting changes efficiency when treatment capacity is limited.

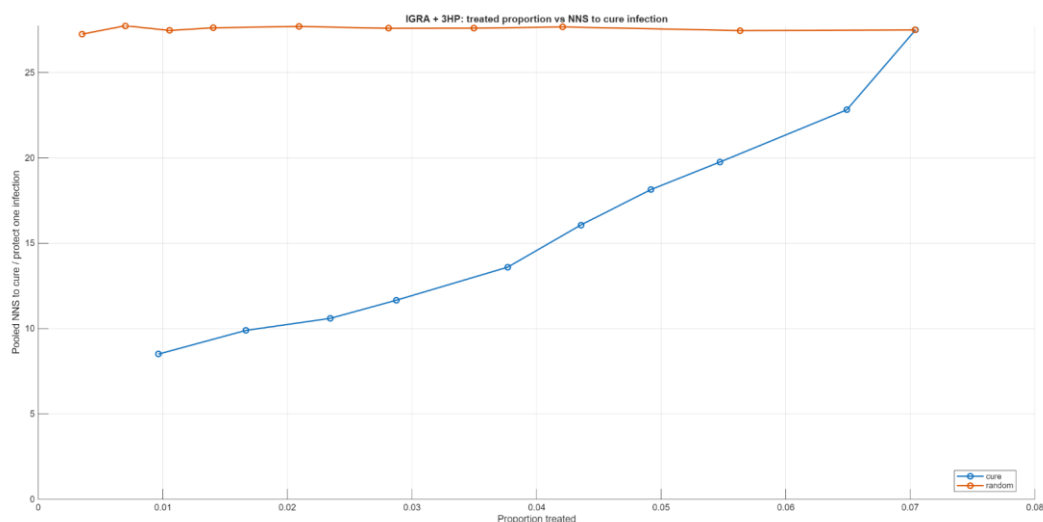


Figure 2. Treated proportion versus number needed to screen (NNS) to protect one LTBI (IGRA + 3HP). Source: chart generated from the current default pathway.

What the figure suggests

- The number needed to screen to protect one LTBI is much lower under cure-targeted testing at low coverage.
- This is the clearest illustration of why targeting matters most when programme capacity is constrained.
- Once treatment reaches a large share of the eligible population, the efficiency advantage narrows.

Results: prevention of future active TB

The prevention-targeted strategy ranks people by predicted preventable active TB risk rather than LTBI risk alone.

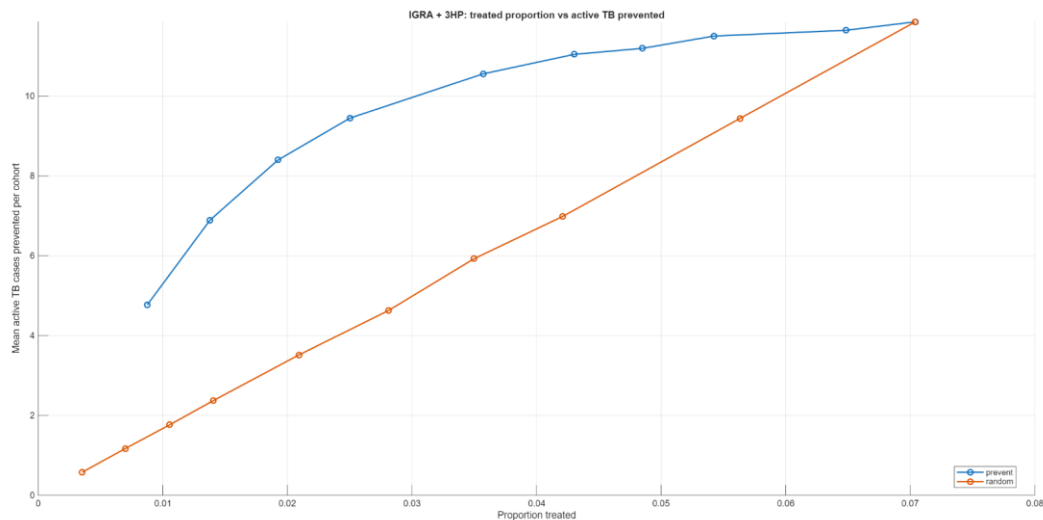


Figure 3. Treated proportion versus active TB cases prevented per cohort (IGRA + 3HP). Source: chart generated from the current default pathway.

What the figure suggests

- The prevention-targeted strategy prevents substantially more future active TB cases than random testing at the same treated proportion.
- The early separation between the curves suggests that the model is successfully identifying people whose combination of LTBI risk and progression risk makes them especially important to reach first.
- As coverage expands, the curves move closer together.

Results: NNS to prevent one active TB case

This is often the most practical efficiency figure for programme planning because it translates future benefit into screening effort.

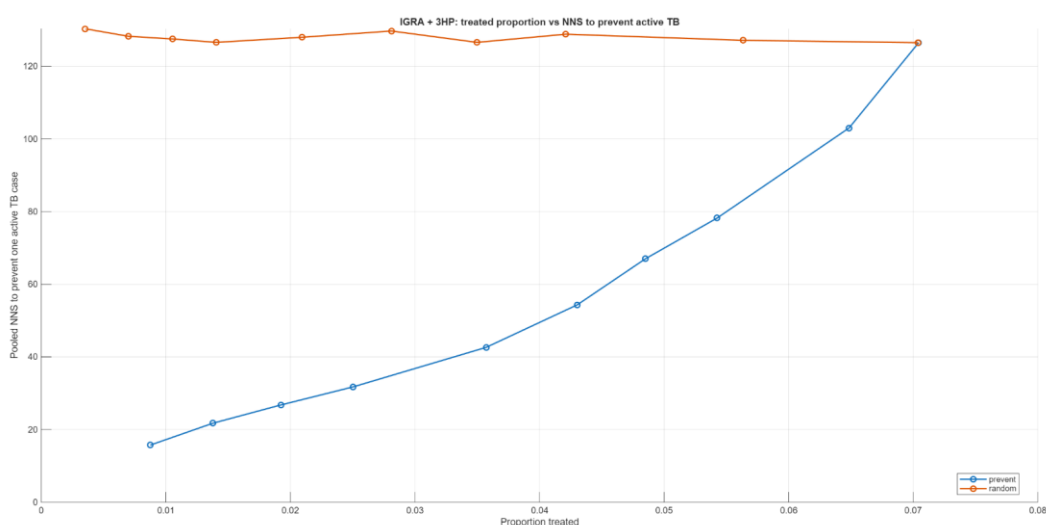


Figure 4. Treated proportion versus number needed to screen (NNS) to prevent one active TB case (IGRA + 3HP). Source: chart generated from the current default pathway. Prevented TB in this build is direct person-level prevention only.

What the figure suggests

- The number needed to screen to prevent one active TB case is dramatically lower under prevention-targeting at low treated proportions.
- This is an important planning result because it translates programme benefit into screening effort.
- The efficiency advantage is largest early and then narrows as more of the reachable population is treated.

Results: LTBI cure / protection

Illustrative APY default scenario using IGRA + 3HP. Targeted and random strategies are shown at the same treated proportion.

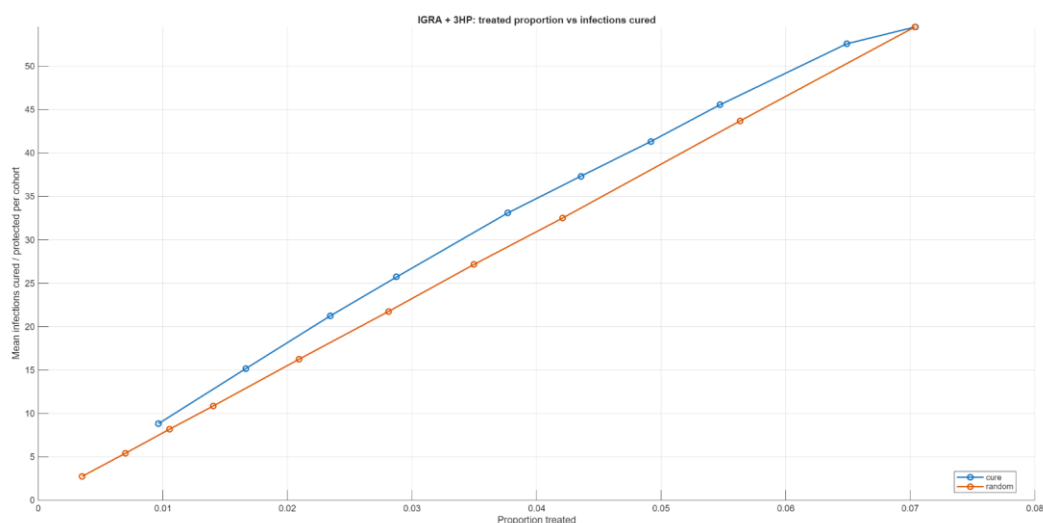


Figure 1. Treated proportion versus infections protected or cured per cohort (IGRA + 3HP). Source: chart generated from the stakeholder-selected default IGRA + 3HP pathway.

What the figure suggests

- At the same treated proportion, the cure-targeted strategy yields slightly more infections protected or cured than random testing across most of the curve
- This is much less impressive than the other efficiency gains
- Once a person is known to have a positive IGRA, there little further efficiency gained through targeting from the perspective of TB infection elimination
- The small gap is due to the slight increase in false positive IGRA results in unselected populations. This would be much more marked if TST were used as the diagnostic

Interpretation for South Australia Public Health

The updated model continues to support targeted testing as a sensible planning strategy in the APY context when programmatic capacity is limited.

The practical implication is not that some people should be excluded. Instead, it suggests that the first rounds of outreach, invitation, and follow-up should be weighted toward those with the highest predicted benefit.

This can be done without compromising equity if the pathway remains open to anyone who presents, is clinically indicated, or requests screening or treatment.

The new model build makes it easier to revisit assumptions over time, because prevalence targets, age distribution, risk-factor prevalence, test choice, treatment choice, and cascade values can now be changed without rewriting the model. The same backend is now also being prepared for a future browser-facing interface, but that interface is not yet part of this report.

Equity and operational guardrails

- Targeting should be used to prioritise early outreach and testing, not to create a rigid eligibility threshold.
- Programmes may reasonably trade away some theoretical efficiency if a simpler workflow is easier to communicate, implement, and sustain.
- The model can now be refreshed more easily when local completion, ADR, or prevalence data improve.

Additional outputs are now available in the current build even though they are not the focus of the four main figures in this report. These include an explicit do-nothing comparator and progression-only attributable-risk outputs for the main reactivation factors. Those outputs are useful for the next round of programme framing and sensitivity analysis.

Conclusions and recommendations

Conclusion

In the current default APY scenario, targeted testing remains the more efficient approach when compared with random testing at the same treated proportion. The gain is greatest when treatment capacity is limited and the programme must decide who to approach first. As more of the reachable population is treated, the efficiency gap narrows.

Recommended policy position

Recommendation

- A targeted strategy should be pursued, weighing the efficiency gains against the convenience and operational simplicity of targeting, while ensuring that no one is refused screening or treatment on the basis of this work.

Recommended actions

- Use targeted testing to plan the order of outreach and early treatment offer when capacity is limited, targeting contacts, people who use marijuana, those with renal impairment, and close contacts. People with multiple morbidities should also be part of the target group.
- Keep access open: no person should be refused screening or treatment on the basis of this modelling work.
- Treat the current results as a planning aid, not a deterministic rule.
-

Model considerations

- Use the model user-input script to refresh results whenever APY-specific screening, completion, or adverse-event data are updated.
- Use the do-nothing and attributable-risk outputs in future rounds to help frame the baseline burden and the progression factors that matter most for prevention planning.
- Reassess the targeting rules periodically against local feasibility, community preference, and implementation experience.

Selected references and source notes

- [1] Barry SE, Brigham M, Gill C, Storken A, Kelly M, Mitnick CD. Tuberculosis Infection in a Remote Aboriginal Australian Community: Prevalence, Predictors, and Public Health Implications. Preprint supplied for this project, 2025.
- [2] Internal MATLAB model files supplied for this project, including tb_screening_mc_model_v9.m and associated v9 add-on scripts.
- [3] Appendix A reproduces the current base-case default_data.csv inputs used in the analytical build. Appendix B reproduces the default_age_distribution.csv inputs used for exact-age sampling.
- Appendix A reproduces the current default_data.csv tables used in the default analytical build. Appendix B reproduces the APY age-distribution table used for exact-age sampling from default_age_distribution.csv. Appendix C provides a practical user manual for running the MATLAB scripts.

Appendix A. Base-case inputs reproduced from default_data.csv

This appendix reproduces the current base-case input tables used by the model. In the v9 build, these values can be replaced through the user-facing runner where appropriate. The browser-facing interface under development is expected to use the same default inputs and scenario structure, but it is not part of this report.

Table A1. Infection-related inputs and retained infection ORs

Input	0-4	5-14	>=15	Disease OR	Infection OR
MJ use	0	5%	30%	3.0	3.65
Close contact	13%	13%	13%	5.0	2.53
Renal disease	0	1%	15%	10.0	2.19

Table A2. Age-specific baseline LTBI risk, progression factors, and population fractions

Input	0-4	5-14	>=15	Disease OR
Baseline LTBI risk by age	1.5%	2.5%	8.0%	-
Diabetes	0	0	30%	3.0
Smoking	0	10%	60%	2.0
Chronic lung disease	0	10%	50%	3.0
Alcohol/drugs	0	40%	40%	3.0
Population fraction	9%	17%	74%	-

Appendix A. Base-case inputs reproduced from default_data.csv

Table A3. Overall prevalence values

Factor	Prevalence
Female sex	56%
Close contact with infectious TB case	13%
Current smoker	52%
Ever smoker	53%
Never smoker	46%
Ever water-pipe marijuana use	18%
Renal disease	12%
Diabetes	25%
Chronic lung disease	19%
BCG vaccinated	68%

Table A4. Comorbidity-count distribution

Parameter	Value
Number of comorbidities = 0	25%
Number of comorbidities = 1	10%
Number of comorbidities = 2	30%
Number of comorbidities = >=3	35%

Appendix B. APY age distribution used for exact-age sampling

The exact-age population in the current build is sampled from the APY age-distribution file supplied for this project, now named default_age_distribution.csv. The smoothed proportion column is used as the default sampling weight. Within each selected 5-year band, a single exact age is drawn uniformly.

Table B1. Age distribution used in the exact-age population generator

Age group	ABS proportion	Smoothed proportion
0-4	10.7%	10.7%
5-9	11.0%	11.0%
10-14	11.0%	11.0%
15-19	9.1%	9.7%
20-24	9.1%	8.6%
25-29	7.4%	8.0%
30-34	7.4%	6.7%
35-39	5.3%	6.0%
40-44	5.3%	5.2%
45-49	5.1%	5.2%
50-54	5.1%	4.7%
55-59	3.9%	4.2%
60-64	3.9%	2.9%
65-69	1.1%	2.0%
70-74	1.1%	1.1%
75-79	1.1%	1.1%

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Age group	ABS proportion	Smoothed proportion
80-84	1.1%	1.1%
85+	1.1%	0.6%

Appendix C. Step-by-step MATLAB user manual

This appendix is a practical guide for South Australia Public Health staff or collaborators who want to rerun the MATLAB model using either the default APY scenario or bespoke local inputs. A browser-facing interface is under development, but until it is available the MATLAB workflow remains the operational route for reproducing results.

Table C1. Main files and their purpose

File	Purpose
tb_screening_mc_model_v9.m	Core model engine
run_tb_screening_user_options_v9.m	Main file for bespoke user inputs
run_tb_screening_example_v9.m	Simple demonstration run using defaults
default_data.csv	Default risk-factor and prevalence inputs
default_age_distribution.csv	Default age distribution for exact-age sampling
v9 add-on scripts	Charts, targeting profile, do-nothing, and attributable-risk summaries

1. Keep the core files together

- Place the following files in the same working folder unless you have a deliberate alternative folder structure:
- tb_screening_mc_model_v9.m
- run_tb_screening_user_options_v9.m
- default_data.csv
- default_age_distribution.csv
- and any v9 add-on scripts you plan to use.

```
results = run_tb_screening_user_options_v9;
```

2. Use the user-facing runner

- Open run_tb_screening_user_options_v9.m.
- The editable section begins at the block marked USER INPUTS.
- Leave a field as [] if you want the current model default to be used.

3. Choose the core scenario settings

- At minimum, choose the cohort size, number of Monte Carlo replicates, screen window, follow-up horizon, screening coverage, and screening strategy.
- The screeningStrategy options are 'random', 'ltbi', 'cure', and 'prevent'.
- For routine planning, use more than one replicate; the illustrative default is 2,000 replicates.

Targeted LTBI screening in the APY Lands - updated planning report

```
user.N = 1500;  
user.nReps = 2000;  
user.screenWindow = 2;  
user.followHorizon = 20;  
user.screenCoverage = 0.30;  
user.screeningStrategy = 'prevent';
```

4. Set or keep the calibration targets

- If you want the APY defaults, leave ltbiPrevalence and activeTbPrevalence empty.
- If you are testing a different setting, enter the desired calibration targets directly.

```
user.ltbiPrevalence = [];  
user.activeTbPrevalence = [];  
% Example alternative  
% user.ltbiPrevalence = 0.10;  
% user.activeTbPrevalence = 0.008;
```

5. Set the age distribution

- The current build can read an age distribution file or an age distribution table.
- If default_age_distribution.csv is stored next to default_data.csv, leaving ageDistributionFile blank will allow the script to auto-detect it.
- The age85PlusMax setting controls how the open-ended 85+ band is handled; the current default is 89.

```
user.ageDistributionFile = '';  
user.ageDistributionTable = table();  
user.age85PlusMax = 89;
```

6. Enter risk-factor prevalence overrides if needed

- For MJ, contact, renal, diabetes, smoking, chronic lung disease, and alcohol, you can enter:
- [] to use default_data.csv,
- a single scalar to use the same prevalence in all 3 age groups, or
- [a b c] to use separate prevalences for 0-4, 5-14, and >=15.
- For female and BCG, use a single overall scalar.

```
user.riskPrev.contact = [];  
user.riskPrev.smoking = [0 0.10 0.60];  
user.riskPrev.alcohol = [0 0.40 0.40];  
user.riskPrev.BCG = 0.68;
```

7. Choose the test and treatment pathway

- Choose IGRA or TST.
- Choose 3HP, 4R, 3HR, 6H, or 9H.
- You can override treatment start, completion, ADR stop, and efficacy if you wish.

```
user.testType = 'IGRA';  
user.regimen = '3HP';  
user.pStartTPT = [];  
user.regimenPComplete = [];  
user.regimenADRstop = [];  
user.regimenEffFull = [];
```

8. Run the model

- Save the file and run the script in MATLAB.
- The script prints a concise summary and returns a results structure.

```
results = run_tb_screening_user_options_v9;
```

9. Use the add-on scripts

- The v9 add-on bundle can generate strategy comparisons, targeting gradients, optima tables, IGRA charts, a do-nothing summary, and attributable-risk outputs.

```
comparison = run_tb_screening_compare_strategies_v9;  
gradient    = run_tb_screening_targeted_gradient_v9;  
profile     = run_tb_screening_targeting_profile_v9;  
optima      = run_tb_screening_targeting_optima_v9;  
charts      = run_tb_screening_igra_charts_v9;  
dn          = run_tb_screening_do_nothing_v9(results);  
attr        = run_tb_screening_reactivation_attributable_v9(results, 50000);
```

10. Interpret the main outputs

- nCuredInfection and nPreventedActiveTB are the most important programme-effect outputs.
- NNS and NNT summarise efficiency.
- The do-nothing script summarises the untreated baseline burden.
- The attributable-risk script isolates the progression-side contribution of each risk factor while keeping infection risk fixed.

11. Troubleshooting

- If MATLAB appears to be using an older script version, run clear functions and rehash, then confirm the path with which <filename> -all.
- If you want a quick demonstration without editing inputs, run run_tb_screening_example_v9.m.
- Keep the default_data.csv and default_age_distribution.csv files aligned with the narrative and charts you plan to use in reports.

```
clear functions
rehash
which run_tb_screening_user_options_v9 -all
```

Final user note

- If MATLAB appears to be holding an old version of a script in memory, run clear functions and rehash before rerunning the workflow.
- Use the user-options runner for bespoke scenarios and keep the example runner unchanged as a stable demonstration file.